

PostgreSQL Setup Guide

Jadara Project

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PostgreSQL 16.14 on Ubuntu 24.04

Contents

1	Installation	2
1.1	Install PostgreSQL	2
1.2	Verify Installation	2
1.3	Check Installed Packages	2
2	Service Management	2
2.1	Start / Stop / Restart	2
2.2	Check Status	2
2.3	Enable on Boot	2
2.4	Check Listening Port	3
3	Configuration	3
3.1	Configuration File Locations	3
3.2	Key Settings in postgresql.conf	3
3.3	Authentication (pg_hba.conf)	3
4	Database & User Creation	3
4.1	Step 1: Connect as PostgreSQL Superuser	3
4.2	Step 2: Create a New User	4
4.3	Step 3: Create the Database	4
4.4	Step 4: Grant Privileges	4
4.5	Step 5: Verify Setup	4
4.6	Step 6: Test Connection	4
4.7	Complete SQL Script	5
5	Backup & Restore	5
5.1	Backup a Single Database	5
5.2	Backup All Databases	5
5.3	Restore a Database	5
5.4	Scheduled Backup with Cron	5
6	Useful Commands	5
6.1	psql Meta-Commands	5
6.2	Useful SQL Queries	5
6.3	Connection Information	6
6.4	Connection String Format	6
7	Quick Reference Card	7

Installation

Install PostgreSQL

```
1 sudo apt update
2 sudo apt install postgresql postgresql-client
```

Verify Installation

```
1 psql --version
```

Expected output:

```
1 psql (PostgreSQL) 16.14 (Ubuntu 16.14-0ubuntu0.24.04.1)
```

Check Installed Packages

```
1 dpkg -l | grep postgresql
```

Key packages:

- postgresql – Main server package
- postgresql-16 – Server binaries
- postgresql-client-16 – Client tools (psql)
- postgresql-common – Cluster manager

Service Management

PostgreSQL runs as a systemd service on Ubuntu.

Start / Stop / Restart

```
1 sudo systemctl start postgresql
2 sudo systemctl stop postgresql
3 sudo systemctl restart postgresql
4 sudo systemctl reload postgresql
```

Check Status

```
1 sudo systemctl status postgresql
```

Expected output (running):

```
1 * postgresql.service - PostgreSQL RDBMS
2   Loaded: loaded
3   Active: active (exited) since ...
4   Main PID: ...
```

Enable on Boot

```
1 sudo systemctl enable postgresql
2 sudo systemctl disable postgresql
```

Check Listening Port

```
1 sudo ss -tlnp | grep 5432
```

Configuration

Configuration File Locations

File	Purpose
/etc/postgresql/16/main/postgresql.conf	Main server config
/etc/postgresql/16/main/pg_hba.conf	Client authentication
/etc/postgresql/16/main/pg_ident.conf	User name mapping

Table 1: PostgreSQL configuration files

Key Settings in postgresql.conf

```
1 sudo nano /etc/postgresql/16/main/postgresql.conf
```

Common settings:

```
1 listen_addresses = 'localhost'
2 port = 5432
3 max_connections = 100
4 shared_buffers = '256MB'
5 work_mem = '4MB'
```

Authentication (pg_hba.conf)

```
1 sudo nano /etc/postgresql/16/main/pg_hba.conf
```

Local connection (default):

	TYPE	DATABASE	USER	ADDRESS	METHOD
2	local	all	postgres		peer
3	local	all	all		scram-sha-256

Note

After changing `pg_hba.conf`, reload PostgreSQL:

```
1 sudo systemctl reload postgresql
```

Database & User Creation

Step 1: Connect as PostgreSQL Superuser

```
1 sudo -u postgres psql
```

This opens the PostgreSQL prompt:

Step 2: Create a New User

```
1 CREATE USER jadara_user WITH PASSWORD 'your_password_here';
```

Replace `your_password_here` with your actual password.

Tip

To generate a random 64-character password in the terminal (outside psql):

```
1 openssl rand -hex 32
```

This produces a 64-character hexadecimal string.

Step 3: Create the Database

```
1 CREATE DATABASE jadara;
```

Step 4: Grant Privileges

```
1 GRANT ALL PRIVILEGES ON DATABASE jadara TO jadara_user;
2 ALTER USER jadara_user CREATEDB;
```

Connect to the jadara database and grant schema permissions:

```
1 \c jadara
2 GRANT ALL ON SCHEMA public TO jadara_user;
3 GRANT ALL ON ALL TABLES IN SCHEMA public TO jadara_user;
4 GRANT ALL ON ALL SEQUENCES IN SCHEMA public TO jadara_user;
5 ALTER DEFAULT PRIVILEGES IN SCHEMA public GRANT ALL ON TABLES TO
   jadara_user;
6 ALTER DEFAULT PRIVILEGES IN SCHEMA public GRANT ALL ON SEQUENCES TO
   jadara_user;
```

Note

The `ALTER DEFAULT PRIVILEGES` commands ensure that tables and sequences created by the postgres superuser during prisma migrate are automatically accessible to `jadara_user`. Without these, Prisma migrations may fail with permission errors on the shadow database.

Step 5: Verify Setup

```
1 SELECT current_user;
2 \q
```

Step 6: Test Connection

```
1 psql -U jadara_user -d jadara -h localhost
```

Complete SQL Script

Here is the full script to run inside psql as the postgres user:

```
1 CREATE USER jadara_user WITH PASSWORD 'your_password_here';
2 CREATE DATABASE jadara;
3 GRANT ALL PRIVILEGES ON DATABASE jadara TO jadara_user;
4 ALTER USER jadara_user CREATEDB;
5 \c jadara
6 GRANT ALL ON SCHEMA public TO jadara_user;
7 GRANT ALL ON ALL TABLES IN SCHEMA public TO jadara_user;
8 GRANT ALL ON ALL SEQUENCES IN SCHEMA public TO jadara_user;
9 ALTER DEFAULT PRIVILEGES IN SCHEMA public GRANT ALL ON TABLES TO
   jadara_user;
10 ALTER DEFAULT PRIVILEGES IN SCHEMA public GRANT ALL ON SEQUENCES TO
   jadara_user;
```

Backup & Restore

Backup a Single Database

```
1 pg_dump -U jadara_user -h localhost jadara > jadara_backup.sql
2 pg_dump -U jadara_user -h localhost -Fc jadara > jadara_backup.dump
3 sudo -u postgres pg_dump jadara > jadara_backup.sql
```

Backup All Databases

```
1 sudo -u postgres pg_dumpall > full_backup.sql
2 sudo -u postgres pg_dumpall --globals-only > globals.sql
```

Restore a Database

```
1 psql -U jadara_user -d jadara -h localhost -f jadara_backup.sql
2 pg_restore -U jadara_user -d jadara -h localhost jadara_backup.dump
3 sudo -u postgres psql -f full_backup.sql
```

Scheduled Backup with Cron

```
1 crontab -e
```

Warning

Always test your backups by restoring them to a test database!

Useful Commands

psql Meta-Commands

Useful SQL Queries

Command	Description
\l	List all databases
\c dbname	Connect to database
\c	Show current connection
\du	List all users/roles
\dt	List tables in schema
\ds	List sequences
\di	List indexes
\dv	List views
\dn	List schemas
\df	List functions
\d tablename	Describe table structure
\dp	List table privileges
\x	Toggle expanded output
\timing	Toggle query timing
\i filename	Execute SQL from file
\o filename	Pipe output to file
\q	Quit psql

Table 2: psql meta-commands

```

1 SELECT current_database();
2 SELECT current_user;
3 SELECT tablename FROM pg_tables WHERE schemaname = 'public';
4 SELECT pg_size_pretty(pg_total_relation_size('table_name'));
5 SELECT pg_size_pretty(pg_database_size('jadara'));
6 SELECT * FROM pg_stat_activity;
7 SELECT pid, now() - pg_stat_activity.query_start AS duration, query, state
8 FROM pg_stat_activity
9 WHERE state != 'idle'
10 ORDER BY duration DESC;

```

Connection Information

Parameter	Value
Host	localhost (or 127.0.0.1)
Port	5432
Database	jadara
User	jadara_user

Table 3: Connection parameters

Connection String Format

```

1 postgresql://user:password@host:port/database

```

Quick Reference Card

Installation

```
sudo apt install postgresql
psql --version
```

Connect

```
sudo -u postgres psql
psql -U user -d db -h localhost
```

Service

```
sudo systemctl start postgresql
sudo systemctl stop postgresql
sudo systemctl status postgresql
sudo systemctl restart postgresql
```

User & DB

```
CREATE USER name WITH PASSWORD
    'pass';
CREATE DATABASE dbname;
GRANT ALL ON DATABASE dbname TO name;
ALTER USER name CREATEDB;
\c dbname
GRANT ALL ON SCHEMA public TO name;
GRANT ALL ON ALL TABLES IN SCHEMA
    public TO name;
GRANT ALL ON ALL SEQUENCES IN SCHEMA
    public TO name;
ALTER DEFAULT PRIVILEGES IN SCHEMA
    public
    GRANT ALL ON TABLES TO name;
ALTER DEFAULT PRIVILEGES IN SCHEMA
    public
    GRANT ALL ON SEQUENCES TO name;
```

psql Commands

```
\l          -- list databases
\du         -- list users
\c dbname   -- connect to db
\dt         -- list tables
\d tablename -- describe table
\q         -- quit
```

Backup

```
pg_dump db > backup.sql
pg_dump -Fc db > backup.dump
pg_dumpall > full_backup.sql
psql -f backup.sql db
pg_restore -d db backup.dump
```

Note

Password placeholder: Throughout this guide, `your_password_here` represents the actual password for `jadara_user`. Generate one with: `openssl rand -hex 32`